

Month 3 Plan – Advanced Chat Features

FastAPI + React + Tailwind | Media, Calls & Notifications

Month 3 Goal

Transform the chat application into a full-featured modern messaging platform by adding file sharing, voice/video calling, notifications, and scalable real-time architecture.

Daily Time Investment (6 Hours)

- 1 Backend (FastAPI): 3.5 hours
- 2 Frontend (React + Tailwind): 2 hours
- 3 Testing & refactoring: 0.5 hour

Week 1 – Media & File Sharing

- 1 Image, video & document upload APIs
- 2 Cloud storage integration (AWS S3 / Cloudinary)
- 3 Secure file access with signed URLs
- 4 Media preview UI (images, videos, PDFs)

Week 2 – Voice & Video Calling

- 1 WebRTC fundamentals & signaling server
- 2 One-to-one audio calling
- 3 Video calling with camera controls
- 4 Call UI (incoming, ongoing, end)

Week 3 – Notifications & Location

- 1 Real-time in-app notifications
- 2 Push notifications (Firebase)
- 3 Live location sharing
- 4 Unread message count & badges

Week 4 – Performance & Scaling

- 1 Redis for WebSocket & presence management
- 2 Rate limiting & abuse prevention
- 3 Optimize DB queries & indexes
- 4 Load testing & profiling

Outcome After Month 3

By the end of Month 3, your project will match real-world chat applications and clearly demonstrate advanced backend, real-time, and system design skills expected from a full-time developer.